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DESIGNING THE TROPICAL SKYSCRAPER

Ken Yeang advocates the careful consideration of climatic factors in designing tall buildings for the tropical countries, and illustrates a new genre of builtform: the Tropical Skyscraper.

Climate is one of the endemic aspects of place over others, and is adopted by some architects as a starting point for design. Principles of design with climate have not been developed for the tall building. This article discusses the adaptation of design to climate for the tall building.

1. THE ABSENCE OF ADEQUATE DESIGN CRITERIA

Although the principles of designing with climate are relatively advanced for low-rise and medium-rise buildings, attention and research have yet to be directed to the tall building type or 'skyscraper'. The need for greater attention to this field is evident. The scale and bulk of the high-rise by far exceed any traditional precedent. The tall building is a novel building type with new servicing systems which requires its own set of design premises even though the fundamental principles of designing with climate remain unchanged.

Historically, tall buildings were structures that symbolized religious expressions or imperial monumentality. However in the present day, high-rise buildings have substituted their religious or imperial associations with those of a commercial commodity. We might compare the high-rise building with the 747 aircraft. Like the airplane, it has become an international piece of technology which every national economy owns. (Even some of the poorest countries in the world have their own national fleet of 747s).

In most instances, national economies of poorer countries do not have the financial capability for a *tabula rasa* to start anew with relocated urban centres of low-rise or medium-rise buildings. Furthermore, the discarding of the existent extensive investment of infrastructures would be wasteful of resources, if the present urban centres were to be replaced. Pressures of increased land values, urban accessibility, expanding urban populations, globalization of national economies and locational preferences of businesses makes the tall building inevitable. What should be of concern is the way these are designed and whether there are effective planning controls for their location, urban design and builtform height.

Architects faced with having to design intensive developments in dense urban contexts, obviously require the criteria for the design of this building type. Unfortunately, there have been no immediately available precedents. If we were to review the historical precedents, we would find that not even the largest traditional or colonial buildings (where these exist) could provide the models for the new tall building type.

It has to be recognised that there is a need for directed research in this field. The fast pace of urban development, particularly in many countries in the hot-humid belt, makes the need for adequate models for designing the high-rise with climate even more pressing.

2. THE TALL BUILDING TYPOLOGY

We might ask what is the tall building? It can be regarded primarily as an intensification of built space over a small site area (or over a small built footprint). This building type permits more usable floor-space to go higher, to make more cash from the land, put more goods, more people, more rents in one place. The high-rise might be seen as a 'wealth-creating' mechanism operating in an urban economy.

We might better define the skyscraper as a multi-storey building generally constructed using a structural frame, provided with high-speed elevators and combining extraordinary height with ordinary room-spaces that would be found in low buildings. The CTBUH (Council of Tall Buildings and Urban Habitat) defines tall buildings as of ten-storeys or more because that is the cut-off height for fire-fighting from ladders in New York City.

Economically, the skyscraper derives from high land values and these are related to urban accessibility, which in turn is a product of road and rail services. Skyscrapers are creations that result from their optimizing land costs and building economics, locational preferences of its occupants, the desire for 'flagship' status by their owners induced by the assertive image associated with the high-rise, and ingenious feats and inventions of architectural and engineering design.

The highrise is also a culmination of a number of building inventions which made it possible: the structural frame and wind-bracing, new methods of making foundations, high-speed elevators, air-conditioning, flush-toilets, large panes of glazing and window-framing, advanced telecommunications and electronics, advanced indoor lighting, ventilation and cleaning technologies.

In seeking to optimise the land-area, the tall building's economics seek to have the maximum internal area on each floor (net areas), and the maximum gross building area for its site (i.e. maximum plot-ratios and minimum 'net-to-gross' ratios). In order to achieve these economic objectives, the following criteria are critical:

- Minimum external wall thickness
- Minimum vertical support size
- Minimum horizontal support thickness
- Minimum vertical circulation/service core area
- Minimum floor-to-floor height

The cost justifications for these are obvious. Minimum external wall thickness and reduced vertical support sizes (i.e. column sizes) and efficient core-areas would increase the net usable floor areas per typical floor. The minimum horizontal support thickness and floor-to-floor heights would lower structural costs and the area of external cladding and hence construction costs.

Clearly, whether in the case of the low-rise or the high-rise building, we find that in those instances where building economics are given precedence over the aesthetic, human or poetic aspects of architectural design, the builtforms that result inevitably end up as bland, inarticulate and plain boxes.

3. THE JUSTIFICATIONS FOR DESIGNING WITH CLIMATE

Is there a need to design with climate? Designing the tall building to take advantage of climate means inevitably some physical and economic departure, to a greater or lesser extent, from those preceding economic optimizing criteria. For instance, the need for sun-shading would increase the thickness of the external wall; or the use of external lift-cores may be less efficient than the central-core layout. What justifications might we have for this departure, besides reasons of architectural aesthetics?

The most obvious justification must be the lowering of costs as a result of lowering the energy consumption in the operation of the building. This can be by as much as 40 per cent of the overall life cycle energy costs of the building, since the bulk of its cost lies in its operational phase. Significant savings in operational costs would justify incorporation of climatically-responsive design features, despite higher initial capital construction costs.

Another rationale is from the impact on the users of the tall building. The climatically-responsive tall building would enhance its users' aesthetic well-being while enabling them to be aware of and to experience the external climate of the place. The climatically-responsive design would provide the building's users with the opportunity to experience the external environment (and the diurnal and seasonal changes where existent), and even the blandness of spending their working hours over a significant pan of their day in an artificial environment that remains constant throughout the year.

A further justification is an ecological one. Designing with climate results in the reduction of the overall energy consumption of the building by the use of passive structural devices (i.e. non-mechanical). Savings in the operational costs means less use of electrical energy resources which are usually derived from the burning of non-renewable fossil-fuels. The lowering of energy consumption further reduces the overall emission of waste heat thereby lowering the overall heat-island effect on the locality.

There is also a regionalist justification for designing with climate. Viewed in the overall perspective of human history and built settlements, the climate of a place is the single most constant factor in our landscape, apart from its basic geological structure. Whilst socio-economic and political conditions may change almost unrecognisably over a period of say 100 years, as may visual tastes and aesthetic sensibilities, its climate remains more or less unchanged in its cyclical course. History shows us that with accumulated human experience and imagination, the architecture of the shelter evolved into diversified solutions to meet the challenges of widely varying climates, indicating that the ancients recognised regional adaptation as an essential principle of architecture. The climatically-responsive building can be seen as having a greater fit with its geographical context.

4. PROBLEMS OF BUILDING IN THE HOT-HUMID CLIMATE

The hot-humid climatic zone is characterised by high ambient temperatures (21-32°C), high humidity, high and fairly evenly distributed rainfall, small diurnal and annual variations of temperature, little seasonal variation, light winds, and long periods of still air. The reflected radiation from the ground is usually low, as the vegetation is dense and the damp soil is dark.

The physiological thermal requirements and hence the building characteristics are the same for the whole year, as the seasonal climatic variations are slight. The main cause of discomfort is the subjective feeling of skin wetness. Continuous ventilation is therefore required to ensure a sweat evaporation rate sufficient to maintain thermal equilibrium and minimum sweat accumulation of the skin. Radiation solar heat gain should be prevented.

The hot-humid zones present two problems to constructors:

- avoidance of excessive solar radiation, and
- provision for moisture evaporation by breezes.

To cope with these, the built structures and settlements need to be built to allow free air movement. The roofs need to be insulated and provided with large overhangs to protect against sun and rain and to provide shade.

Traditional houses in this climatic zone have floors which are raised to keep them dry and to allow for air to circulate underneath. In urban planning, the preference is for 'chequer-board' spacing of buildings to enhance air-circulation.

Under hot conditions, the thermal controls in the building should:

- prevent heat gain
- maximize heatloss
- remove any excess heat by mechanical cooling.

The first two objectives can be achieved by means of 'microclimatic control' through site-layout and internal space-planning, controlling and planning air-movements, external wall and space orientation (such as configuring the building and selective placement of the various functional parts of the building) and the use of structural and constructional passive means of control.

The last objective above can be achieved only by means of an active energy input or by mechanical means of control.

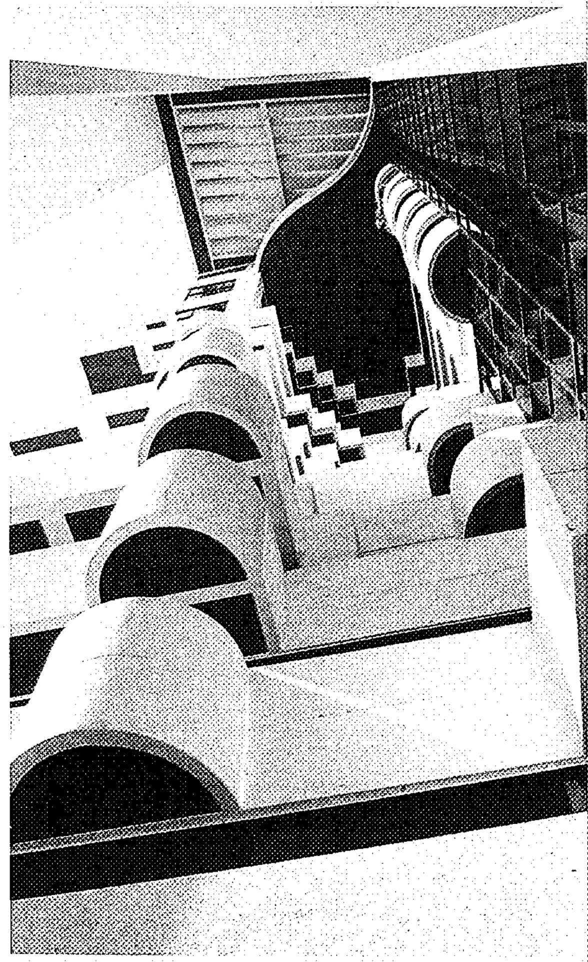
5. THE IDEA OF THE TROPICAL HIGH-RISE

The idea of the 'tropical high-rise', simply stated, is a tall building whose builtform's design and planning have responses to and take advantage of the climatic factors of the locality. The resultant builtforms might even be regarded as providing alternatives to the bland glass 'box' which does not acknowledge the climate of the place and could be built internationally anywhere.

Adapting the fundamental principles of designing with climate for the low-rise and medium-rise buildings, and then re-examining these in aggregate with those design and planning principles for the tall building, we would arrive at a number of design principles. These are as follows:

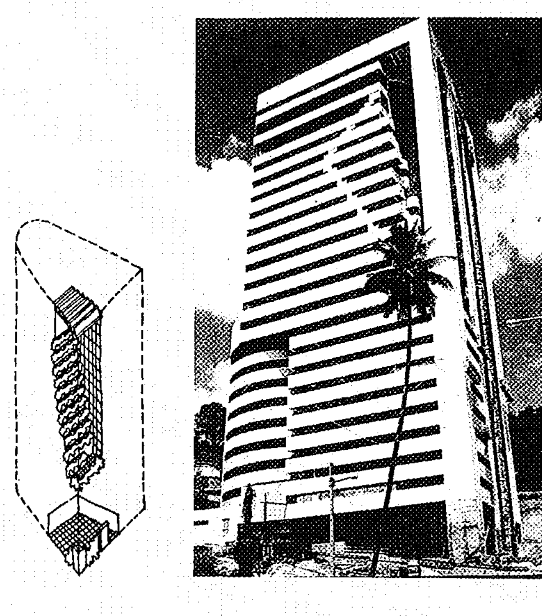
SERVICES CORE POSITION

The position of the services core is a key

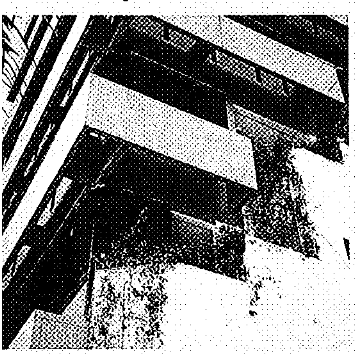


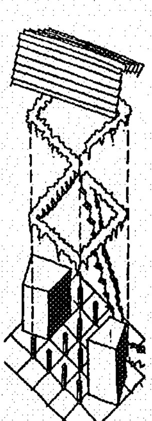
PLAZA ATRIUM, Kuala Lumpur
(T.R. Hamzah & Yeang Sdn, Bhd).

- *Recessed glazing on window walls facing east and west sides; curtain-wall glazing is used inside atrium space.*
- *A large multi-storey atrium is used as transitional 'in-between' space and as wind-scoop.*
- *Terraces for planting and office users face the atrium space.*
- *A louvred roof over atrium space lets in diffused light and lets out hot-air from atrium.*
- *Naturally ventilated ground floor ('open-to-sky' space).*
- *Naturally ventilated and sun lit lavatories, stairways and lift-lobbies.*



IBM PLAZA, Kuala Lumpur
(T.R. Hamzah & Yeang Sdn, Bhd).
- *Floor plan oriented north-south at an angle to the geometry of the site.*
- *Recessed and shaded windows to east and west sides.*
- *Service cores at east and west sides.*
- *Vertical landscaping from green area at ground level and linked continuously up the faces of the building combined with balconies.*
- *Louvred roof over last three floors.*
- *Naturally ventilated ground floor ('open-to-sky' space).*
- *Naturally ventilated and sun lit toilets, stairways and lift-lobbies.*





consideration in the planning of the high-rise building's floor-plate and in determining its configuration. Besides their obvious structural ramifications, services-cores determine which parts of the building's peripheral walls will be openings and which external walls, as well as affecting its thermal performance and orienting its views to the outside. Essentially there are three possible types of core positions: 'central-core', 'double-core' and 'single-sided core'.

In the hot-humid climates, the solar path is generally east-west. Consequently the services-cores should preferably be located either on the 'hot' east, or the west side.

In the case of the double-core configuration, the cores are positioned peripherally to the floor-plate. This configuration offers many benefits: the two cores can be located on the 'hot' sides and can provide 'buffer-zones' to the internal spaces. Studies have shown that savings in air-conditioning loads result from a double-core configuration with window openings running north-south and cores on the east and west sides. This placement of the cores would meet the first two objectives above – preventing heat gain into those user-spaces which require greater cooling than the core areas, and providing a buffer (as a form of 'spatial' thermal insulation) to the hot sides of the building. This would maximise heat loss away from the user-spaces.

NATURAL VENTILATION AND SUNLIGHT TO SERVICES-CORES

The services-core are parts of the tall building which contain the lift shafts, lift lobbies, main and escape stairways, riser-ducts, toilets and other service rooms. These zones should have natural ventilation, sunlight and a good view to the outside wherever possible. This principle dictates a peripheral rather than a central-core position. If placed externally, further energy-savings can be achieved through reduced requirements for mechanical ventilation, artificial lighting and the need for mechanical pressurization ducts for fire-protection.

By providing a pleasant view from these areas, the users of the building can see out and experience the natural environment as soon as they exit from the lift-car and can have the opportunity to experience natural sunlight and ventilation. In contrast to the central-core situation, the building user leaves the lift-car only to enter into an artificially-lit upper floor lobby or passageway.

BUILDING ORIENTATION

Tall buildings are exposed more directly than others to the full impacts of external temperatures and direct sunlight. So their overall orientation has an important bearing on energy conservation. The greatest source of heat gain in the hot-humid climate can be the solar radiation entering through the window. The direct radiation transmitted varies markedly with the time of day and the angle of incidence, remaining fairly steady until about 50°C and dropping sharply at 60°C.

Main openings facing north-south minimizes solar insulation and thus the building's air-conditioning load. If the site does not align with the sun's geometry on it's east-west path, other built-elements such as the services-cores can follow the geometry of the site to optimize column grids, basement car-parking layouts, etc.

Unless important views lie elsewhere, windows should face the direction of the least direct solar insulation i.e. on the north and south sides, in conjunction with curtain-walling if this is deemed aesthetically desirable.

Some builtform shaping adjustments or shading devices (e.g. balconies) may need to be provided for those site locations which lie further north or south of the Tropics. Generally the window openings should orientate north-south unless important views require other orientations or openings.

If required for aesthetic reasons, glazed curtain-wall may be used on the non-solar facing façades. But on the 'hot' east and west sides, some form of solar-shading is required making due allowance for glare and the quality of light entering the spaces. The west wall has the highest intensity at the hottest time of the day.

RECESSES AND SKY-COURT

One means of reducing the solar heat gain through windows is by shading the 'hot' sides of the building using deep recesses in the external wall. These may take the form of totally recessed windows, balconies or small-scale courtyards in the upper floors or 'sky-courts'.

Besides providing shade, these sky-courts could have full-height glazed doors leading to the recessed terrace areas giving the office users the possibility of access to the spaces. These sky-courts thus created could act as emergency evacuation spaces, as terraces for planting and landscaping, as flexible zones to permit the future addition of executive wash-rooms or kitchen spaces, and as a semi-enclosed transitional space for the building users to experience and enjoy the view of the external environment.

ATRIA, AIR-SPACES AND WIND-SCOOPS

The use of multi-storey recessed 'transitional spaces' would represent another way of shading the 'hot' sides of a tall building. Positioned either centrally or peripherally, such huge transitional air-spaces or atria perform the same traditional role – between inside and outside – as did the verandas in vernacular architecture. The building users with access to such spaces have the option of experiencing the external natural environment under semi-enclosed conditions.

Such spaces need not be totally enclosed from above: the atrium might be shielded by a 'louvred roof' and solar-shading to encourage wind-flow to the inner areas of the building. These devices may even be extended over the entire face of the building to create a multi-storey atrium that would act as wind-scoops to direct natural ventilation to the inner parts of the building as well as for the exit of hot air resulting from the Venturi effect. By organising the building's internal passageways to be perpendicular to the atrium space and to lead off this space, they can act as 'conduits' for ventilation and breezes into the internal office space. The airflow can be controlled by adjustable louvres at the openings.

ADJUSTABLE OPENINGS

The external walls of the building should be regarded more as a 'sieve' rather than as a sealed skin. These should function as a permeable membrane with adjustable openings. Ideally the tall building's external walls should act like a filter that has variable parts to control good cross-ventilation, provide solar protection, regulation of wind-swept rain and discharge of heavy rain.

The roof design of the tall building is of less importance than in the low-rise and medium-rise building, since the external wall area of the tall building by far exceeds its roof area. Furthermore, the height of the building makes any overhangs of the roof effective only for the upper few floors. However, the direct solar-heat structural absorption of the roof in the last floors needs to be considered. In the case of the tall building, much of the roof is usually occupied by mechanical plants which offer some insulation.

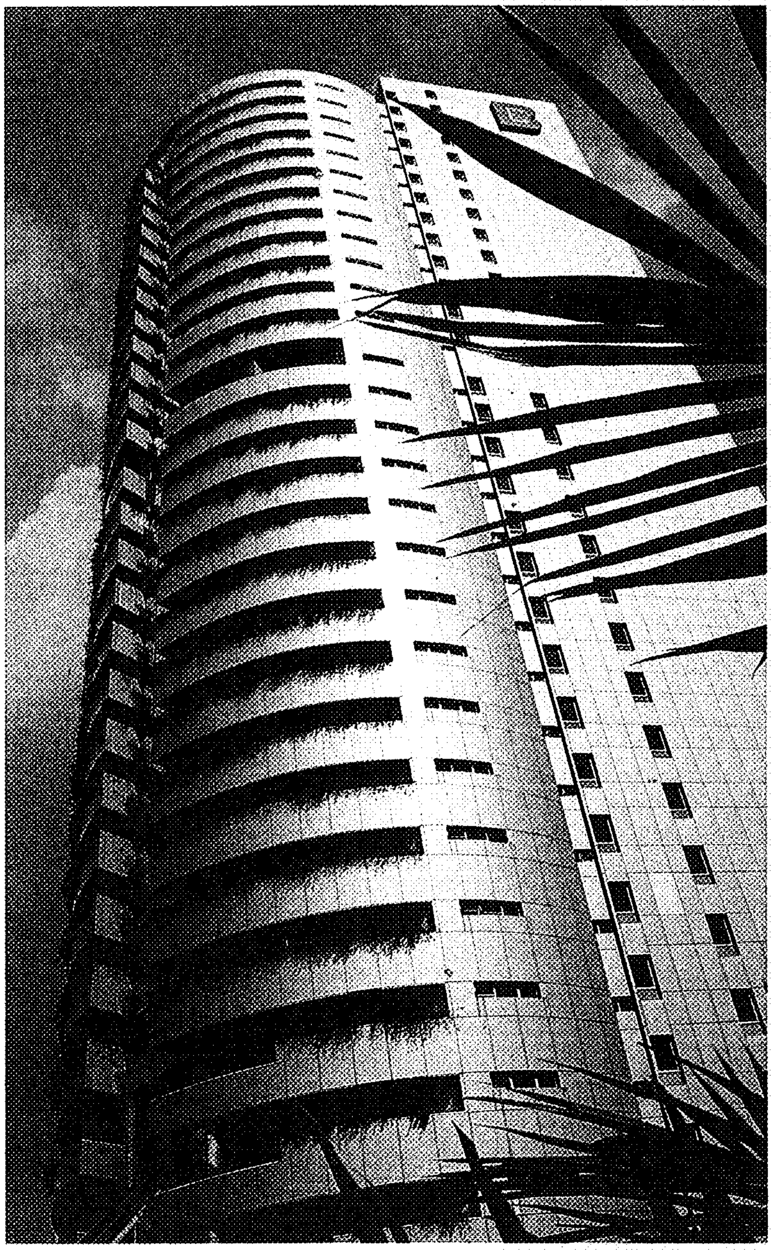
For good natural cross-ventilation to be effective (say 30 air-changes per hour) as a substitute for air-conditioning, the best arrangement for windows is full wall-openings on both the windward and leeward sides, with adjustable or closing devices which can assist in channelling the airflow in the required direction following the change of wind. Ideally, openings should be as large as possible. However, the high velocity of winds at the upper floors of the tall building could make this impractical. A recessed window with means of adjusting the through-flow of wind and wind-swept rain would provide an alternative to air-conditioning when desired.

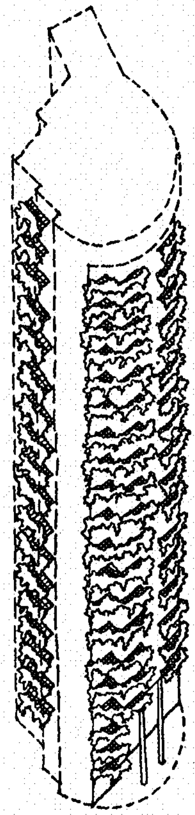
THE OPEN GROUND FLOOR

The ground floor of the tall building deserves special consideration. It should be entirely open to the outside as a naturally-ventilating space. It should not be enclosed nor air-conditioned so that it can effectively act as a transitional space between the hot outsides and the cooler insides. Care should be taken, however, to keep out wind-swept rain and wind-turbulences at the base of the building.

The relation of the ground floor to the street is also very important. In many parts of the world, the sealed indoor air-conditioned atrium marked the demise of street-life.

Free-standing buildings further increase alienation from the street. They reduce the pedestrian communication and movement into and around buildings from the street resulting in the classic 'island site' situation.





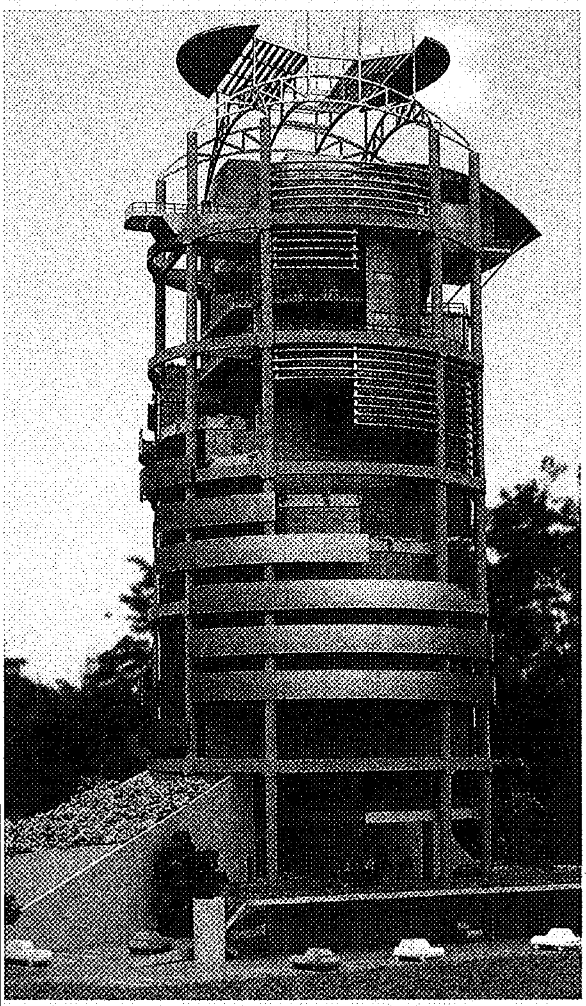
VERTICAL LANDSCAPING

Although soft-landscaping strategies for low- and medium- rise buildings are relatively well developed (common solutions include using planter-boxes and roof-planting), they are notably less advanced in the case of tall buildings. The introduction of planting and soft-landscaping to the face of the tall building or integrated with its inner sky-courts in the building's upper parts can have aesthetic and ecological benefits as well as providing effective climatic responses.

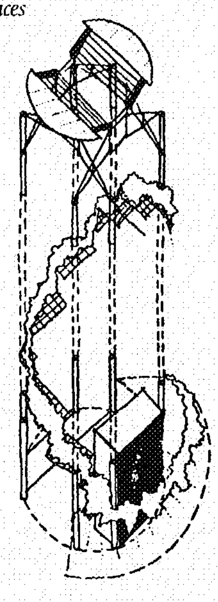
The use of tropical planting enhances the aesthetics of the tall building as a tropical structure. However, the planting, besides providing shading to the internal spaces and to the external wall, also minimizes heat reflection and glare into the building. The evaporation processes of the plants can also be an effective cooling device to the face of the building.

There is an ecological justification to the introducing of planting to the tall building. In any ecosystem, climate is held to be the predominant influence even though other biotic

MENARA BOUSTEAD, Kuala Lumpur
(T.R. Hamzah & Yeang Sdn, Bhd).
- *Balconies located at corners of building with planting and full height sliding glazed-doors for natural ventilation to inner spaces (if required).*
- *Ventilated double-cladding external wall system using composite aluminium panels.*
- *Service cores on the east and west sides*
- *Naturally ventilated and sunlit toilets, stairways and lift-lobbies.*
- *Naturally ventilated ground floor ('open-to-the-sky' space).*
- *Recessed windows on hot sides but curtain-wall glazing to north side of façade.*



MENARA MESINIAGA, Kuala Lumpur
(T.R. Hamzah & Yeang Sdn, Bhd).
- *Spiralling vertical landscaping that climbs up the face of the building.*
- *Ramped turfing and planting up the lower floors of the building.*
- *Recessed and shaded windows on east and west sides.*
- *Curtain-wall glazing to north and south sides.*
- *Single-core services core on hot side.*
- *Naturally ventilated and sunlit toilets, stairways and lift-lobbies.*
- *Spiralling balconies on the external wall with full height sliding doors for natural ventilation (if required) to the interior offices and as transitional spaces for building users.*
- *Sun-shaded roof over the top floor.*



factors such as flora and fauna with the soils would also have an influence on the system. In most urban locations all that remains of the site's earlier ecosystem components is probably the topsoil with the upper geological strata, and a much simplified and reduced fauna complement. In any new built system, we should recognise the fundamental ecological value of increasing the site's biological diversity.

Physical continuity between planting is important for encouraging species diversity. A variety of different indices have been used by ecologists to describe the 'health' of an ecosystem and, whilst it should be recognised that these sometimes contradict one another ('one man's flower being another man's weed'), most see a high level of species diversity as desirable.

To achieve physical continuity in 'vertical landscaping' in the tall building, we can have a system of linked and stepped planter-boxes organised as 'continuous planting zones' up the faces of the building. These would then permit some extent of species-interaction and migration to take place and be linked to the ecosystem at the ground level.

The alternative option would be to separate the planting into unconnected boxes. However this can lead to species homogeneity which necessitates external inputs (regular human maintenance) to remain ecologically stable.

Of the three possible relationships between built systems and planting – 'juxtapositioning', 'intermixing' and 'integration', – integration is clearly preferred.

CONFIGURING THE FLOOR LAYOUT

The design of the tall building's floor layout besides responding to commercial considerations of the building, should take account of human modalities and local cultural patterns of living and working, of privacy and community, all of which had developed in relation to the locality's climate.

These should be reflected in the building's configuration, in its floor-depth, the positioning of its entrances and exits, the provision for human movement through and between spaces, its orientation and in its external views. The building plan should provide humanity, interest and scale, as well as allowing for air-movement through spaces, the provision of sunlight internally, adequate ventilation, and so forth.

RELATIONSHIP TO URBAN CONTEXT

The tall building has three levels of physical and climatic relationships with its urban context: firstly, that of the city as a whole, its image and systems; secondly at the city-block level; and lastly at the level of the local pedestrian scale and existing patterns of life around, inside and through the building's lower floors.

Generally any erection of new building on a vacant site increases the site's overall density of inorganic mass. The environmental impact of this at the city-level is compounded by that of its servicing systems, power, lighting, air-conditioning, waste output, and so on which not only consume energy in their operations, but may also make polluting discharges. All these need to be addressed by passive climatic-control systems which would reduce energy consumption and the reduction and control of polluting systems.

At the city-level, the overall energy exchange of the city – seen holistically as an ecosystem – is increased with the introduction of any new tall building. Depending upon the shading and wind-flow, the new tall building may lead to 'heat-island' effects and other micro-climatic problems. This needs to be counteracted by the introduction of vertical landscaping, the use of heat-sink claddings and the reduction of air-conditioning uses which result in heat discharge from condensers.

At the city-block level, it will affect the wind-flow around adjoining buildings at both the upper levels as well as at the lower levels and wind-tunnel tests should be carried out early on to ascertain the impact on the site and on the surrounds. The heat emitted by the building systems would also affect the micro-climate of the locality. The introduction of vertical landscaping would increase the biotic components in the local ecosystem.

It is clear that urban planning in locating tall buildings affects several environmental factors which are important to the comfort and well-being of the inhabitants. Careful consideration should therefore be given to the pattern of sun and shade; the degree of protection from radiation, rain and wind: the ventilation conditions. These in turn, will be influenced by: the dimensions and particularly the heights of the buildings; the spacing of the buildings; the variation in heights in any one section of a town; the orientation of the street network and the distribution and extent of open spaces and gardens. In a warm, humid climate, planning should be directed towards optimization of ventilation conditions and providing the maximum protection from solar radiation.

EXTERNAL MATERIALS AND THERMAL INSULATION

Roofs and walls in the tall building should be constructed of low-thermal-capacity materials with reflective outside surfaces where these are not shaded. The roof should be of double construction, be provided with a reflective upper surface and also a ceiling of highly reflective upper surface. The use of a good thermal insulation layer is recommended.

The thermal forces acting on the outside of the tall building are a combination of radiation and convection impacts besides others. The radiation component consists of incident solar radiation and of radiant heat exchange with the surroundings. The convective heat impact is a function of exchange with the external air and may be accelerated by air movement. The exchange effect may be increased by diluting the radiation over a larger area by the use of curved surfaces such as vaults, domes, atrias, louvred or irregular-surfaced roofs, which will at the same time increase the rate of convection transfer.

Wall surfaces having direct solar insulation (especially the east and west sides) should be shaded. Their cladding material's insulation and 'time-lag' characteristics should be taken into consideration. External materials used might be those that are effective heat-sinks (e.g. aluminium composites) or be designed to have a 'double layered' ventilating space.

Another very effective protection against radiation impact, especially in the external wall design, is the selective absorptivity and emissivity characteristic of a material, especially under hot conditions. Materials which reflect rather than absorb radiation, and which release the absorbed heat as thermal radiation more readily, bring about lower temperatures within the building.

6. DISCUSSION

The well-tried traditional precedents of building have enabled their occupants in the past to be comfortable with maximum economy. However, these traditional solutions should be studied but certainly not copied since they have developed under conditions and building intensities that have changed considerably through time.

It is not practical to design a building exclusively on economic, functional or on stylistic grounds alone and then expect a few minor adjustments to give a good indoor climate. Unless a design is fundamentally correct in all aspects, no subsequent specialist can make it function satisfactorily.

The location's climate must be taken into account at the design stage when deciding on the overall concept of a project, on the layout and orientation of buildings, on the shape and character of structures, on the spaces to be enclosed, and on the spaces allowed between buildings.

The various propositions here for the design of the high-rise in the hot-humid climate provide us with a useful starting set of principles. It can also serve as an alternative approach to designing the tall building in the hot-humid climate as an option to the minimalist box that is often equated with the modernist builtform.

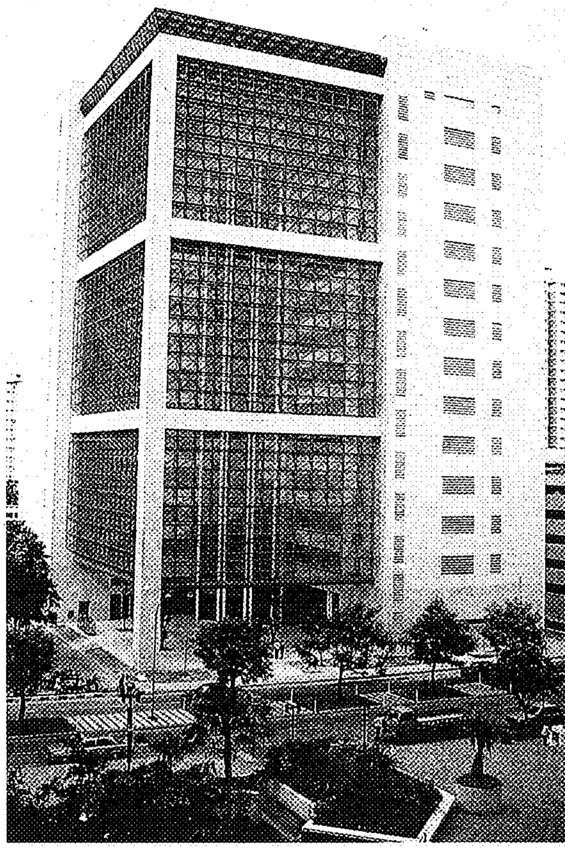
It might be speculated that these factors would point towards a new builtform and type. Detractors might argue that this may lead to another kind of urban monotony replacing the modernist one that is often criticized. But more likely, the wide range of climatic variations and the multitude of individual site characteristic differences would dictate a wide range of variations in building configurations, and the consequences of designing the tall building with climate could well generate a greater range of possibilities in building aesthetics.

The full realization of the built environment of a tall building that is totally naturally ventilated and sunlit would obviously be dependent upon the users themselves accepting a less consistent comfort level of working environment and accepting the inconvenience of having to occasionally manipulate external wall devices to respond to changing climate conditions (unless these are automated).

The reason is that climate is not always consistent. For instances, there may be moments of still air with no through ventilation or the presence of overcast sky. Responding to these may well seem initially troublesome especially when compared to the convenience of the constant artificially controlled internalized environment of an air-conditioned and artificially-lit office. The issue becomes more of a social-economic question dependant upon the level of comfort that is acceptable to the community and the acceptability of its ecological and aesthetic justification.

The idea of a climatically-responsive tall building may well direct a gradual but phased change in user behaviour and preferences away from the present fully artificial working environment to one that is a naturally (or partially naturally) ventilated environment, perhaps supplemented by lower-energy use mechanical means (e.g. electric fans).

Notwithstanding this, tall buildings designed with principles of climate are viable without any lowering of comfort levels. The adoption of these principles is justified on grounds of continual savings both in cost and energy consumption. CM



PARKWAY SERVICE CENTRE,
Singapore *(Akitek Tenggara II)*.
- *Naturally-ventilated atrium space with louvred cladding to atrium walls.*
- *Un-enclosed access balconies to all upper floors that overlook the atrium.*
- *Skylight to atrium for natural lighting.*

MAHAVELI OFFICE COMPLEX,
Colombo *(Geoffrey Bawa)*.
- *Transitional spaces as naturally ventilated 'sky terraces' to upper parts of the tower.*
- *'Thin slab' builtform configuration for natural lighting to office spaces.*

Photographs: T. R. Hamzah & Yeang Sdn. Bhd.

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